Who Wrote This?
Midterm Report

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Abstract

This study quantifies the proliferation of unlabeled AI-generated content across five domains: News, Reddit, Amazon, arXiv, and Resumes. By benchmarking contemporary corpora against pre-2021 baselines using Binoculars and GPTZero, we estimate machine-authored saturation within these sectors. We further analyze sentiment and engagement to characterize the distinct rhetorical signatures and impacts of AI versus human authorship

1 Performed Tasks

1.1 Data Collection

We have successfully initiated data collection across our five target domains, establishing a pre-2022 / preLLM baseline (assumed to be purely human-generated) and a post-2022 modern dataset. For Social Media, rather than relying on standard API scraping, we sourced 80,000 records across seven thematic domains from the Pushshift Reddit archive via Academic Torrents. For Commercial Product Reviews, we have secured a subset of the Amazon Customer Reviews dataset. For Academia, we have queried the arXiv Open Access API to collect all papers published on arxiv in Dec 2021 and Dec 2025. For News, we used the news subset of Common Crawl to randomly sample articles from a set list of domains across both dataset catagories. Lastly, we are actively compiling a dataset of professional summaries and resumes from open-source repositories and public datasets.

1.2 Preprocessing

We designed a standardized, non-destructive preprocessing pipeline with a focus on preserving authorship-relevant signals while removing artifacts that could interfere with detection models. For Reddit, this included removed HTML entities and stripping invisible Unicode characters (e.g., zero-width spaces) that disrupt tokenization. We explicitly removed bare URLs and media embeddings, as they do not carry meaningful authorship information and introduce noise into model inputs.

Importantly, we avoided common normalization steps such as lowercasing, stemming, or punctuation removal, as these surface-level features are critical for distinguishing between human and AI-generated text.

Additional preprocessing steps included filtering out empty or deleted entries, applying length constraints, enforcing English-language detection, and performing exact deduplication. Similar cleaning procedures were applied to news and review datasets. For resumes, we removed personally identifiable information (e.g., names, contact details, education history) to isolate content more likely to reflect generative patterns, such as professional summaries and experience descriptions.

1.3 Initial Testing

Our testing methodology for the *Binoculars* model revolves around its optimal context window. Based on the literature, Binoculars is most effective when evaluating inputs of exactly 256 tokens. For contiguous text such as a standard Reddit post, we sample a single 256-token chunk, padding the input if it is too short, or taking just one sample if it exceeds the limit (assuming a single, contiguous social media post only requires one chunk to reveal its nature). Conversely, for structured or multi-part documents like research papers or news articles, we sample one chunk per section, allowing us to capture varying degrees of "AI-ness" throughout a single document.

2 Potential Risks and Challenges

Risk 1: Binocular Failure The open-source detector we are using appears to perform poorly on positive control datasets.

Risk 2: GPTZero Cost Seemingly performant commercial detectors are prohibitively costly at the dataset scales we are targeting.

3 Risk Mitigation Plan

Mitigation 1: We are conducting further investigation into the causes of poor detector performance. We are also searching for different models, and considering slightly pivoting the analyses focus from human sentiment to AI Model effectiveness.

Mitigation 2: We plan to request research access for GPTZero, which may provide free usage.

4 Individual Contributions

Jad Barmada: Led the design, collection, and preprocessing of the Social Media (Reddit) dataset. Key contributions include:

- **Dataset Engineering:** Sourced 80,000 records from the Pushshift archive with a balanced split (40,000 pre-2021, 40,000 post-2021).
- **Unbiased Sampling:** Implemented Vitter's Algorithm R (Reservoir Sampling) to obtain uniformly random samples and eliminate viral-post bias.
- **Pipeline Design:** Applied temporal stratification across 7 domains and developed cleaning, filtering, and deduplication pipelines.

Mostafa Dewidar: Led the design, collection, and preprocessing of the Academia (arXiv) dataset as well as initial tests. Key contributions include:

- **Organizer:** Organized project workstreams, meetings, and collaborative sessions
- **Dataset Engineering:** Built arXiv dataset (20,000 papers split pre-2021 and post-2025) and preprocessed LaTeX data into $\sim$250-word chunks for model input
- **Control Testing:** Ran initial Binoculars experiments (Dec 2021 vs Dec 2025) showing minimal difference (0.9% vs 1.5%). Then conducted positive control tests using Claude and GPT-4o rewrites. Observed weak detection performance, indicating unreliable detection; currently exploring alternative methods

Conner Nguyen: Led the design, collection, and preprocessing of the Online Reviews (Amazon) dataset. Key contributions include:

- **Dataset Engineering:** Sourced 40,000 reviews (20,000 pre-2022, 20,000 post-2022).
- **Sampling Strategy:** Sampled proportionally across categories to maintain representative distribution.

Anay Mody: Led the design, collection, and preprocessing of the Professional Branding dataset. Key contributions include:

- **Dataset Engineering:** Collected 6,000+ resumes (2,400+ pre-2021, 3,400+ post-2021).
- **Sampling Strategy:** Applied random sampling and structured the dataset into four categories: real, template-based, LLM-generated, and synthetic.

Ryan Angle: Led the design, collection, and preprocessing of the News dataset. Key contributions include:

- **Dataset Engineering:** Collected 40,000 news articles via Common Crawl (20,000 pre-2021, 20,000 post-2021).
- **Sampling & Preprocessing:** Sampled articles from WARC files and applied preprocessing for model compatibility.